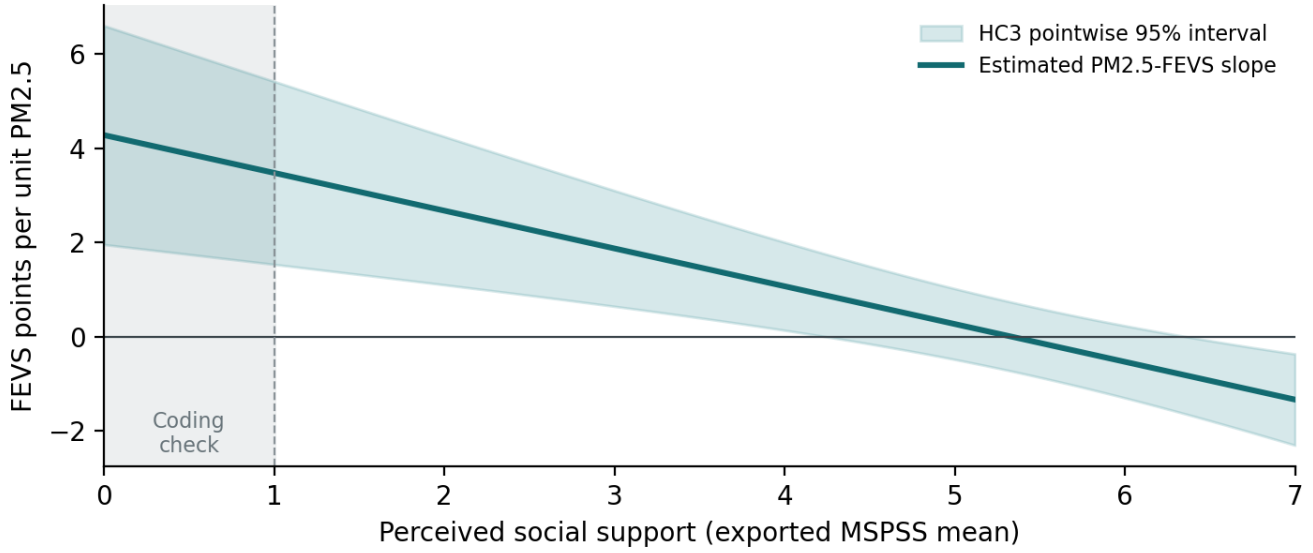
Financial vulnerability and social reward

RF1-SRA / Katie Social Doors | Main results brief | November 3, 2026

PRELIMINARY: financial-score verification and MSPSS coding checks remain open. N=97 and settled exclusions are unchanged.

1. Support Moderates The PM2.5-FEVS Association

Interaction b = **-0.802**; HC3 95% CI **[-1.208, -0.396]**; two-sided p = **0.000168**; Holm p = **0.000336** across the two primary behavioral tests.



The plotted quantity is the estimated PM2.5-FEVS slope, not the FEVS score itself. The association is more positive at lower support and weakens as support increases. Bands are pointwise, not simultaneous. The shaded 0-1 region flags values below the standard MSPSS mean-score range.

Model And Influence

FEVS ~ PM2.5 + MSPSS + PM2.5 x MSPSS + age + gender. N=97; residual df=91. Every single-case omission retains a negative interaction (b=-0.939 to -0.724); maximum Cook's D=0.135. This does not rule out sensitivity to multiple questionable scores together.

Scoring check: five exported MSPSS values are below 1, including one zero. They were retained as supplied, not declared missing or imputed. Verify the administered coding and recompute scores from original items. FEVS/OAFEM versions, item coding, timepoints and missing-item rules also require confirmation.

Eight behavioral models were reported: two primary FEVS/support interactions and six secondary tests. None of the other seven interactions survives its within-family Holm adjustment. Behavioral results and complete tables

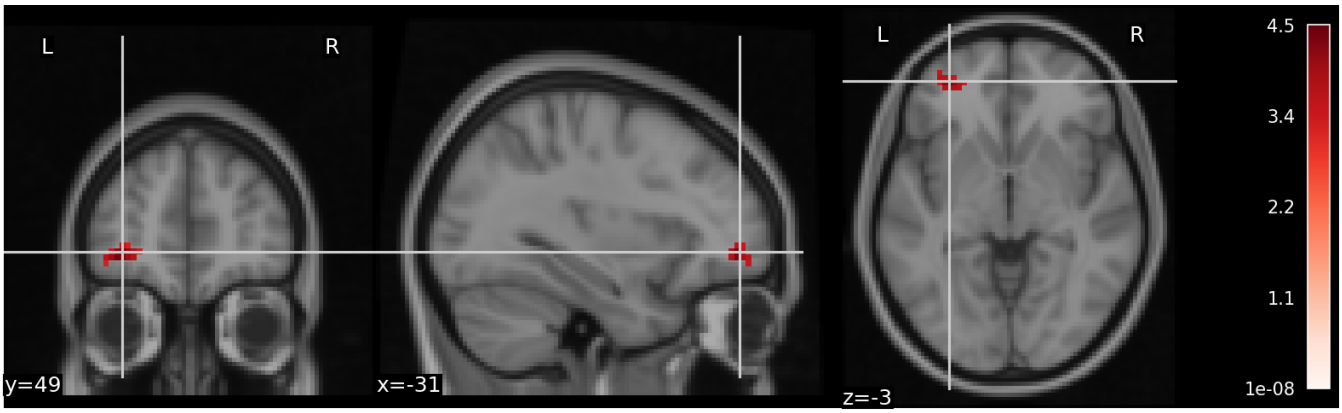
2. FEVS And Social-Specific Winning

ACTIVATION | Social Doors win - nonsocial Doors win

Higher FEVS is associated with a larger social-minus-nonsocial winning activation difference in a small left anterior/ventral frontal region near OFC. This contrast isolates the difference between domains during wins; it is not the domain difference in win-minus-loss responding.

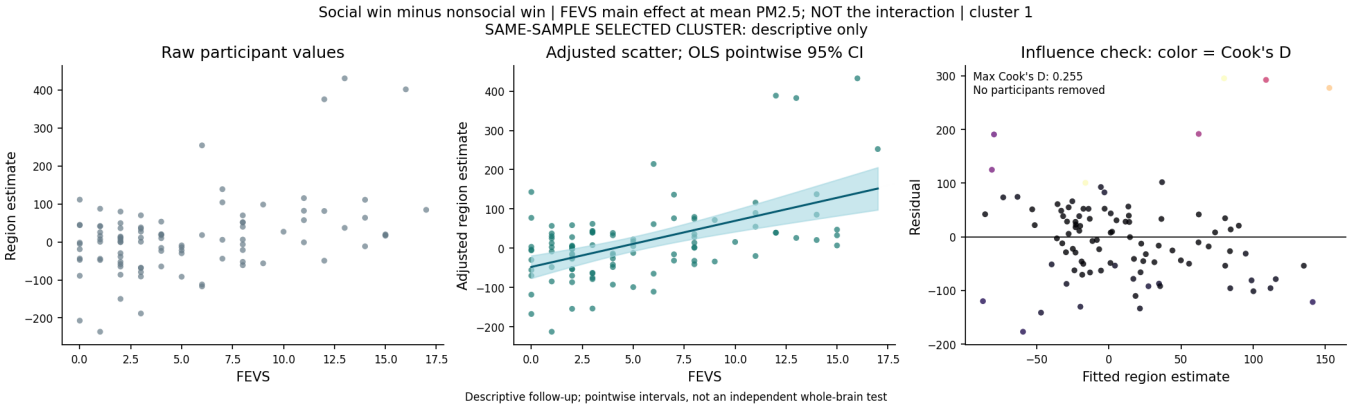
FEVS coefficient	Voxels	Cluster-FWE p	Peak MNI (mm)
Separate FEVS predictor	56	0.0411	-31.6, 49.5, -6.7
At mean PM2.5	67	0.0315	-31.6, 49.5, -3.7

Social win minus nonsocial win | FEVS main effect at mean PM2.5; NOT the interaction



N=97; positive FEVS coefficient; cluster-FWE p<.05; CDT Z-equivalent 3.1.
Coordinates are MNI mm. Atlas annotations describe location, not independent confirmation.

Displayed: positive FEVS coefficient at mean PM2.5, CDT Z-equivalent 3.1. The model includes FEVS, PM2.5 and their interaction, plus age, gender, scanner era and signed FD/tSNR differences. This is a conditional main effect, **not** a significant winning-related FEVS x PM2.5 interaction.



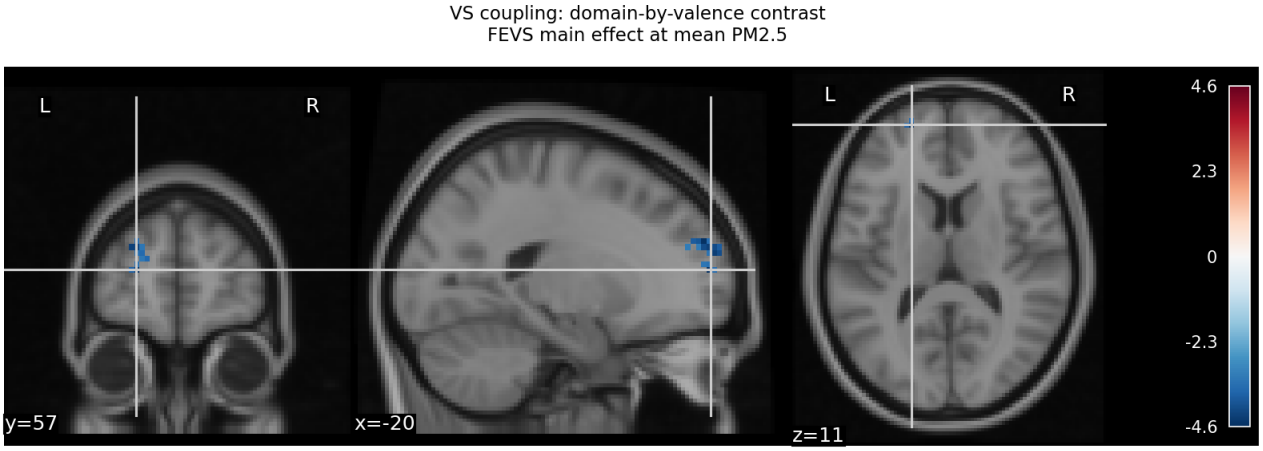
The two related models overlap by 48 voxels (Dice=0.78); they are not independent replications. No single-case omission reverses either descriptive FEVS slope. Selected-cluster scatterplots and their analytical intervals are descriptive, not independent confirmation of the whole-brain test.

Anatomical caution: substantial portions of the clusters are white-matter-labeled in the atlas. Report coordinates and cautious frontal localization; do not call this an established cortical vmPFC or VS activation effect. FEVS activation report

3. FEVS And VS Connectivity

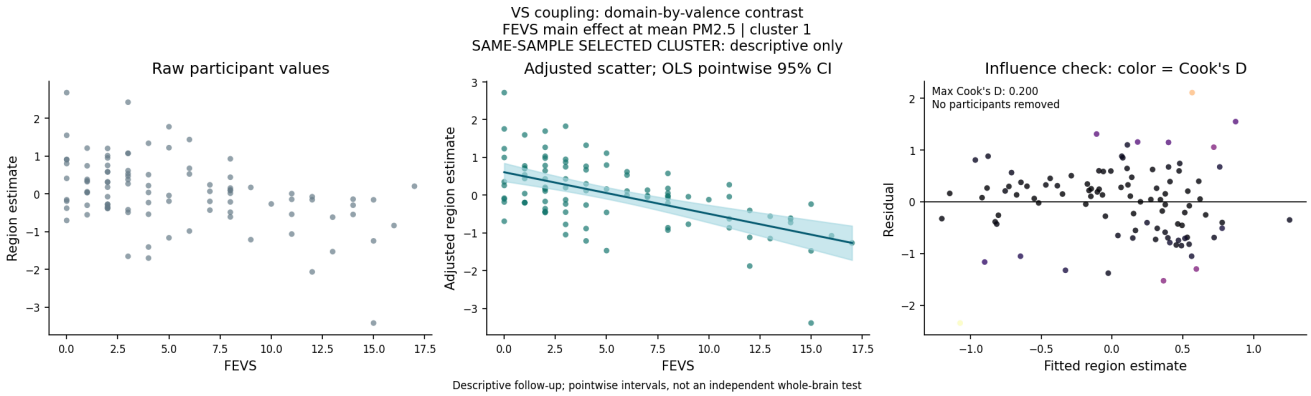
VS PPI | [Social win - social loss] - [nonsocial win - nonsocial loss]

At mean PM2.5, higher FEVS is associated with a smaller social-by-valence difference in task-modulated VS coupling with a left frontal-pole target: **cluster-FWE p=0.0303**, 44 voxels; peak MNI **[-20.8, 57.6, 11.2]**. The frontal cluster is the coupling target, not the VS seed.



N=97; blue = negative coefficient; cluster-FWE p<.05; CDT Z-equivalent 3.1.
MNI coordinates; atlas labels are approximate annotations.

Displayed: negative FEVS coefficient at mean PM2.5, CDT Z-equivalent 3.1. This is not a FEVS x PM2.5 interaction, not winning-specific coupling and not a statement about globally weaker connectivity.



The descriptive slope remains negative under every single-case omission (-0.114 to -0.098). Selected-cluster plots are not independent confirmation. Most cluster voxels are cortex-labeled, but the peak is white-matter-labeled.

Full VS connectivity report

Interpretation And Next Steps

FEVS-FIRST SYNTHESIS | Contextual and neural correlates, not a causal pathway

Financial exploitation vulnerability (FEVS) has both contextual and neural correlates in this older-adult sample. The PM2.5-FEVS association varies with perceived support; separate analyses associate FEVS with social-winning activation and socially differentiated VS coupling. These findings do not establish a connecting causal pathway.

What The Combined Results Do Not Show

There is no matching corrected neural PM2.5 x support interaction, and no corrected FEVS x support interaction. The completed catalog also has no FEVS social-winning VS-connectivity finding and no social-by-valence activation finding. The behavioral and neural results are complementary FEVS associations, not a demonstrated buffering mechanism or mediation pathway.

Analysis Scope

N=97, paired ses-01, approved age-55+ imaging cohort. Neural inputs are the completed no-TEDANA sm-5 FEAT variant and original VS seed. Neural inference uses 10,000 permutations and cluster extent at Z-equivalent CDTs 2.6/3.1; highlighted FEVS clusters pass at 3.1. Cluster-FWE is per map, not across the exploratory model/outcome/CDT family. No three-way interaction or new exclusion is introduced in this brief.

This focused summary presents selected main findings, not the full analysis inventory. Other tested models, nulls and secondary findings remain documented. VS report | Full results and model inventory

Scoring Closeout Before Final Interpretation

Verify the administered MSPSS response coding and calculation for all 97 participants, including the five exported values below 1. The standard mean-score range is 1-7; a documented modified version could differ. Low values do not by themselves prove questionnaires are missing. Obtain the original FEVS/OAFEM item coding, instrument versions, source timepoints and missing-item rules, and recompute totals against the export.

Preserve all 97 participants, original exports and completed neural results. Save any corrections as a versioned export and assess their implications before changing analyses. Do not replace zeros, impute items or exclude participants simply to resolve the coding question.

Status

Computation and output validation are complete. Financial-score and MSPSS coding checks remain open before final interpretation. No participant selection, score correction or new inferential analysis was performed to create this brief.

MSPSS instrument and standard scoring instructions